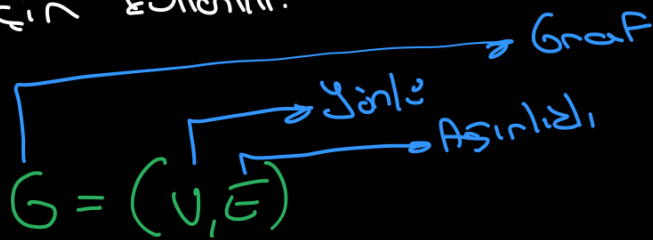


Dinamik Programlama

Floyd Warshall

Tanım = Düzgünler arası en kısa yolun tespiti

için kullanılır:



$l(i, j)$ = i düğümü ile j düğümü arasındaki uzaklık

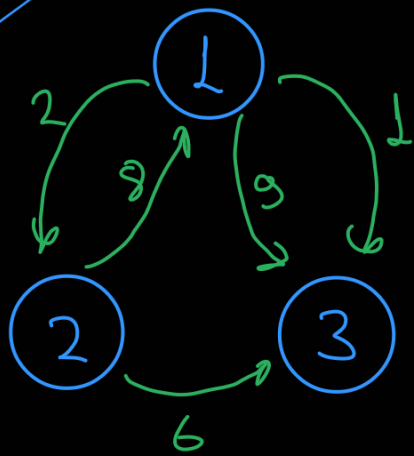
d_{ij}^k = i ile j arası ulaşmada, k bölünecek yer

terimler. Alternatif uçur gibi. Bazen alternatif uçur

da ha uçurdu.

$$d_{ij}^k = \begin{cases} l(i, j) & \text{if } k=0 \\ \min \left\{ d_{ij}^{k-1} \text{ OR } d_{i,k}^{k-1} + d_{k,j}^{k-1} \right\} & \text{if } 1 \leq k \leq n \end{cases}$$

1. Netz



Herberg: bin 2 Sekunden
digerine er kann mehr frey
buhuz?

$$D^0 =$$

	1	2	3
1	0	2	9
2	8	0	6
3	1	∞	0

→ Direkt u. v. u. k. g. b. i. d. u. n
0 → Direkt

→ Direkt g. d. g. k.

$k=1$ iter

$i=1$ ↓ $j=1$

$$D(1,1) = \min(D(1,1), D(1,1) + D(1,1)) = 0$$

$$D(1,2) = \min(D(1,2), D(1,1) + D(1,2)) = 2$$

$$D(1,3) = \min(D(1,3), D(1,1) + D(1,3)) = 9$$

$i=2$

$$D(2,1) = \min(D(2,1), D(2,1) + D(2,1)) = 8$$

$$D(2,2) = 0$$

$$D(2,3) = \min(D(2,3), D(2,1) + D(1,3)) = 6$$

$i=3$ ↓ ↓ $j=1$

$$D(3,1) = \min(D(3,1), D(3,1) + D(1,1)) = 1$$

$$D(3,2) = \min(D(3,2), D(3,1) + D(1,2)) = 3$$

$$D(3,3) = 0$$

$k=2$

$i=1$

$$D(1,1) = 0$$

$$D(1,2) = \min(D(1,2), D(1,1) + D(1,2)) = 2$$

$$D(1,3) = \min(D(1,3), D(1,1) + D(1,3)) = 9$$

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$$D(2,2) = 0$$

$$D(2,3) = \min(D(2,3), D(2,1) + D(1,3)) = 6$$

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$$D(3,3) = 0$$

$k=3$

$i=1$

$$D(1,1) = 0$$

$$D(1,2) = \min(D(1,2), D(1,1) + D(1,2)) = 2$$

$$D(1,3) = \min(D(1,3), D(1,1) + D(1,3)) = 9$$

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$$D(3,2) = \min(D(3,2), D(3,1) + D(1,2)) = 3$$

$$D(3,3) = 0$$

$$\begin{array}{c}
 D^0 \text{ 'in tabiri} \\
 D^1 \text{ der} \\
 D^2 \text{ der}
 \end{array}
 \begin{array}{c}
 D^0 \\
 D^1 \\
 D^2
 \end{array}
 \begin{array}{c}
 \begin{bmatrix} 0 & 2 & 9 \\ 8 & 0 & 6 \\ 1 & \underline{\underline{3}} & 0 \end{bmatrix} \\
 \begin{bmatrix} 0 & 2 & \underline{\underline{8}} \\ 8 & 0 & 6 \\ 1 & 3 & 0 \end{bmatrix} \\
 \begin{bmatrix} 0 & 2 & 8 \\ \underline{\underline{7}} & 0 & 6 \\ 1 & 3 & 0 \end{bmatrix}
 \end{array}$$

En kısa yol maliyetini içeren matris

İşlem Maliyeti

for (k=1 to n)

for (i=1 to n)

for (j=1 to n)

$$D_{i,j} = \min \{ D(i,j), D(i,k) + D(k,j) \}$$

Zaman Maliyeti = $\Theta(n^3)$ (iç içe 3 for)

Bellek Maliyeti = $\Theta(n^2)$ → 3 eleman için 9 verili "2" tablo saklıyoruz bellekte

D^0, D^1, D^2 bu tablo uneninde? değıştirmek sadece

Dinamik programlama Örnekleri

Floyd Warshall

$$G=(V,E)$$

$$l(i,j) = i \text{ ile } j \text{ arasındaki uzaklık}$$

$$d_{i,j}^{(k)} = \begin{cases} l(i,j) & \text{if } k=0 \\ \min \left(d_{i,j}^{(k-1)}, d_{i,k}^{(k-1)} + d_{k,j}^{(k-1)} \right) & \text{if } 1 \leq k \leq n \end{cases}$$

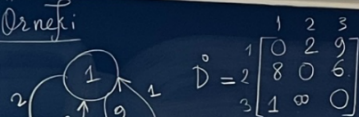
$$D^0 = \begin{bmatrix} 0 & 2 & 8 \\ 7 & 0 & 6 \\ 1 & 3 & 0 \end{bmatrix}$$

$$D^1 = \begin{bmatrix} 0 & 2 & 9 \\ 8 & 0 & 6 \\ 1 & 3 & 0 \end{bmatrix}$$

$$D^2 = \begin{bmatrix} 0 & 2 & 8 \\ 8 & 0 & 6 \\ 1 & 3 & 0 \end{bmatrix}$$

$$D^3 = \begin{bmatrix} 0 & 2 & 8 \\ 7 & 0 & 6 \\ 1 & 3 & 0 \end{bmatrix}$$

for k=1 to n
for i=1 to n
for j=1 to n
 $D(i,j) = \min \{ D(i,j), D(i,k) + D(k,j) \}$



$$D(1,1) = \min(D(1,1), D(1,2) + D(2,1))$$

$$D(1,2) = \min(D(1,2), D(1,3) + D(3,2))$$

$$D(1,3) = \min(D(1,3), D(1,2) + D(2,3))$$

$$D(2,1) = \min(D(2,1), D(2,3) + D(3,1))$$

$$D(2,2) = 0$$

$$D(2,3) = \min(D(2,3), D(2,1) + D(1,3))$$

$$D(3,1) = \min(D(3,1), D(3,2) + D(2,1))$$

$$D(3,2) = \min(D(3,2), D(3,3) + D(3,3))$$

$$D(3,3) = 0$$

$$D(1,1) = 0$$

$$D(1,2) = \min(D(1,2), D(1,3) + D(3,2))$$

$$D(1,3) = \min(D(1,3), D(1,2) + D(2,3))$$

$$D(2,1) = \min(D(2,1), D(2,3) + D(3,1))$$

$$D(2,2) = 0$$

$$D(2,3) = 6$$

$$D(3,1) = \min(D(3,1), D(3,2) + D(2,1))$$

$$D(3,2) = \min(D(3,2), D(3,3) + D(3,3))$$

$$D(3,3) = 0$$

$$D(1,1) = 0$$

$$D(1,2) = \min(D(1,2), D(1,3) + D(3,2))$$

$$D(1,3) = \min(D(1,3), D(1,2) + D(2,3))$$

$$D(2,1) = \min(D(2,1), D(2,3) + D(3,1))$$

$$D(2,2) = 0$$

$$D(2,3) = \min(D(2,3), D(2,1) + D(1,3))$$

$$D(3,1) = \min(D(3,1), D(3,2) + D(2,1))$$

$$D(3,2) = \min(D(3,2), D(3,3) + D(3,3))$$

$$D(3,3) = 0$$

Laman
Tüketimi
 $\Theta(n^3)$

İçerik
Maliyeti
 $\Theta(n^2)$